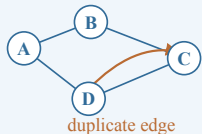


1 Environment

Graph G_t



Active rules R_t

Duplicate validator

Relation validator

Fixed validator registry

2 Observation

14 features $\cdot o_t$



Graph quality

Rule quality

Remaining defects

Budget + active fractions

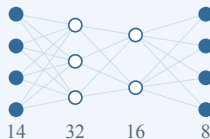
8-action mask $\cdot m_t$



Graph + rule identities

3 RL policy

Double DQN



Online action values

Mask $\rightarrow$ select a_t

ϵ -greedy during training

4 One operation

Graph operations

Remove isolated nodes $\cdot$ deduplicate

Restore relations $\cdot$ remove dangling edges



Rule operations

Acquire deletion / augmentation / both

Prune low-precision validators

R_t

$+r$

R_{t+1}

Update $G, R \rightarrow$ quality gains – costs – new-violation penalty

Next observation and mask are recomputed from the updated environment.

Learning from transitions

Replay: $(o_t, a_t, r_t, o_{t+1}, \text{done}, m_{t+1})$

Reward r_t uses graph and rule quality gains.

Online network θ

Masked selection $\cdot$ Huber loss

Target network θ^-

Evaluate bootstrap action

100-step copy